# Anomaly Detection for Anti-Money Laundering: Literature Review

*Internal research note – January 2026 (v1)*

**Scope.** Survey of ML/DL methods for AML anomaly detection, focusing on 2020–2026. Covers graph-based, deep learning, traditional ML, contextual/KYC, feature engineering, datasets, and emerging paradigms. Not a formal paper—a working reference for project planning.

**Audience.** Data scientist with anomaly detection experience starting an AML project.

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# 1 Graph and Network-Based Approaches

Financial transactions form natural graphs (nodes = accounts, edges = transfers). Graph methods detect laundering patterns invisible in tabular data.

## 1.1 GNNs for AML

The Elliptic dataset (Weber et al., 2019, KDD Workshop) is the foundational benchmark: 203K Bitcoin transactions, 234K edges, ∼2% illicit. GCN achieved ∼0.97 AUC, showing graph structure adds value over node features alone.

Cheng et al. (2024, *Frontiers of Computer Science*, arXiv:2411.05815) survey 100+ GNN methods for financial fraud, finding GNNs substantially outperform traditional approaches.

| **Method** | **Dataset** | **AUC** | **F1 (illicit)** | **Note** |
|---|---|---|---|---|
| Random Forest | Elliptic | 0.96 | 0.47 | Node features only |
| GCN | Elliptic | 0.97 | 0.52 | +graph structure |
| GAT | Elliptic | 0.97 | 0.51 | Attention-based |
| GraphSAGE | Elliptic | 0.97 | 0.54 | Inductive |
| EvolveGCN-H | Elliptic | 0.97 | 0.58 | +temporal |
| ATGAT | Elliptic++ | 0.91 | – | 9.2% over XGBoost |
| CARE-GNN | YelpChi | 0.94 | 0.86 | Anti-camouflage |
| PC-GNN | YelpChi | 0.97 | 0.88 | Imbalanced learning |
| GraphSAGE | AMLSim | 0.85–0.95 | 0.70–0.85 | Varies by pattern |

Key architectures:

- **GCN/GAT/GraphSAGE**: Standard baselines. GraphSAGE preferred for inductive (unseen node) settings. Recent temporal-aware GAT (ATGAT) on Elliptic++ achieves 0.913 AUC.
- **Improved HybridNet** (2025): Combines GraphSAGE + GATv2 + temporal encoding.
- **LineMVGNN** (2025): Line-graph multi-view GNN for directed financial graphs.
- **LayerWeighted-GCN**: Adaptive layer weighting across GNN depths.

## 1.2 Self-Supervised and Contrastive Learning on Graphs

Label scarcity is fundamental in AML (labeled SAR data is rare and sensitive):

- **LaundroGraph** (Cardoso et al., 2022, ECML-PKDD Workshop): Self-supervised graph embeddings → XGBoost. F1 improved ∼5–10% over raw tabular features.
- **Bit-CHetG** (2024): Subgraph contrastive learning on heterogeneous graphs for laundering group detection.
- **HCLNet** (2025): Hypergraph contrastive learning for group-level fraud semantics.
- **Causal-DHG** (2024): Dynamic hypergraph + causal intervention + multi-view contrastive learning.
- **GRAD** (2025, TheWebConf): Guided relation diffusion + supervised graph contrastive learning from limited labels.
- **SpaceGNN** (2025, ICLR): Multi-space projections with distance-aware propagation.

## 1.3 Heterogeneous Graphs

Real financial networks have multiple entity types (persons, accounts, merchants, devices) and relation types:

- **CARE-GNN** (Dou et al., 2020, CIKM): Camouflage-resistant; RL-based relation selector. AUC 0.94 on YelpChi.
- **PC-GNN** (Liu et al., 2021, WWW): Pick-and-choose sampling for imbalance. AUC 0.97.
- **xFraud** (Rao et al., 2022, VLDB): Industrial-scale heterogeneous GNN deployed at eBay scale with explainability. Billions of nodes.
- **HAN**: Hierarchical attention (node-level + meta-path semantic-level). 2–8% AUC improvement over homogeneous GNNs.
- **HTGNN** (2025): Heterogeneous temporal GNN integrating spatial, temporal, and semantic information for real-time detection.

### 1.4 Temporal and Dynamic Graphs

- **EvolveGCN** (Pareja et al., 2020, AAAI): RNN evolves GCN weights over time. F1 0.58 vs. 0.52 for static GCN on Elliptic.
- **TGN** (Rossi et al., 2020): Memory-augmented temporal graph network with per-node memory. Adapted for AML by subsequent works.
- **GTCT** (2025): Graph-Temporal Contrastive Transformer for transactional behavior modeling.
- **Continual Graph Learning for AML** (Deprez et al., 2025, *WIREs Computational Statistics*): Reviews replay-based, regularization-based, and architecture-based strategies against catastrophic forgetting.
- Streaming GNNs from Ant Group, PayPal for real-time processing on evolving graphs.

### 1.5 Subgraph and Community Detection

Money laundering involves coordinated account networks. Node-level detection misses this:

- **FlowScope** (Li et al., 2020, KDD): Dense subgraph detection for scatter-gather laundering patterns. Orders of magnitude faster than subgraph enumeration.
- **SMoTeF** (2024, *Applied Intelligence*): Smurf detection using temporal order and flow analysis. Applied to 180M+ transactions, 31M accounts.
- **GARG-AML** (2025): Scalable, interpretable smurfing detection. Outperforms SOTA on large-scale networks.
- **AMLGaurd** (2025): Unsupervised GAE-based detection via reconstruction discrepancy.
- **Graph Feature Preprocessor** (2024, KDD): Real-time subgraph pattern mining on streaming transactions. Multicore parallelism.
- **The Shape of Money Laundering** (2024): Subgraph representation learning on Elliptic2 revealing laundering scheme shapes (peeling chains, nested services).

## 2 Deep Learning (Non-Graph)

### 2.1 Sequence Models (LSTM, GRU)

Transaction sequences are temporal $\Rightarrow$ RNNs are natural:

- LSTM/GRU outperform SimpleRNN (vanishing gradients). GRU often preferred for AML: fewer params, AUC-ROC $> 0.7$. GRU matches or beats LSTM on most AML benchmarks.
- Attention-enhanced LSTM (Niu et al., 2021): Attention focuses on most suspicious subsequences.
- Jensen and Iosifidis (2023): Replace rule-based triggers with auto-extracted sequential features at bank level.

### 2.2 Transformers

- **Tang and Liu (2024)**: Unified Transformer-BDI with distributed knowledge distillation. F1 92.87%, AUC 96.73%.
- **TREASURE** (KDD 2026): Transformer foundation model for high-volume transaction understanding.
- **TransactionGPT** (2025): GPT-style pretraining on payment data for behavioral anomaly detection.
- **FraudGT** (2024, ICAIF): Graph Transformer for financial fraud.
- **TabTransformer, TransTab, BERT, RoBERTa** applied to structured transaction data (2024).
- **Spatial-Temporal-Aware Graph Transformer** (2024, IEEE TII).

### 2.3 Autoencoders and VAEs

Trained on normal transactions, high reconstruction error = anomaly:

- **VAE for CC Anomaly** (2024, *Big Data Mining and Analytics*): CNN-based VAE; hybrid VAE + classifier models.
- **GAN+VAE Hybrid** (2025): Combines GANs and VAEs for large payment flow anomaly detection.
- **LVAESC-SFD** (2026): Correlation-based feature selection + VAE, 98.24% accuracy.
- **Denoising Autoencoder** (Bundesbank, 2023): Detection + interpretability on financial tabular data.
- **Self-Attention + VAE** (2024, *Expert Systems with Applications*): Fund transfer fraud with attention for long-range dependencies.
- Shallow autoencoders (1–2 layers) often sufficient for tabular AML. AUC ∼0.85–0.90 on CC fraud, ∼0.80–0.85 on AML-specific data.

### 2.4 GANs for AML

Dual purpose: synthetic data generation and direct anomaly detection.

- **WGAN for data augmentation**: Synthetic fraudulent transactions improve F1 by 10–15%.
- **SMOTE-GAN hybrids** (2023): SMOTified-GAN, SMOTE+GAN, GANified-SMOTE variants.
- **Advanced R-GAN** (2024): Regularized GAN with spectral normalization against mode collapse.
- **CTGAN, Copula GAN, TVAE**: State-of-the-art synthetic data tools. 96–99% utility equivalence to production data.
- Alan Turing Institute project (commenced autumn 2024): GAN-based synthetic AML data for UK FCA initiative.

### 2.5 LLMs and Foundation Models

Newest frontier (2025–2026):

- **FLAG** (KDD 2025): LLM-enhanced GNN for fraud detection. Uses large language models to generate semantic node features from raw transaction metadata, which are then fused with structural graph embeddings. Demonstrates that LLM-derived features can substitute for manual feature engineering in graph-based AML pipelines.

- **GuARD** (KDD 2025): Text-rich, graph-informed language model for anomaly detection. Jointly encodes unstructured text fields (e.g., transaction memos, KYC notes) and graph topology into a unified representation. Particularly relevant for AML settings where free-text fields carry signals that tabular features miss.
- **DGP** (AAAI 2026): Dual-granularity prompting with graph-enhanced LLMs. Introduces two levels of prompting—transaction-level and subgraph-level—to capture both local and structural anomaly signals. Achieves competitive performance with GNNs while offering natural-language explanations as a by-product.
- **AuditCopilot** (NeurIPS 2025 Workshop): LLMs for bookkeeping fraud. Applies instruction-tuned LLMs to detect anomalies in accounting records and journal entries. Highlights the potential for LLMs in adjacent financial crime domains beyond transaction monitoring.
- **Co-Investigator AI** (2025): Agentic framework for faster, more accurate SAR generation. Combines retrieval-augmented generation with structured case data to draft SAR narratives, reducing analyst writing time. Positions LLMs as analyst-assistance tools rather than standalone detectors.
- **FinGPT** (IJCAI 2023): Open-source financial LLM. Pre-trained on financial corpora and fine-tunable for downstream tasks including sentiment analysis, risk assessment, and fraud classification. Provides a foundation model baseline for financial NLP tasks, though AML-specific fine-tuning remains underexplored.

# 3 Traditional ML

## 3.1 Gradient Boosting

Dominates tabular AML. Consistent with Grinsztajn et al. (2022, NeurIPS): tree-based models outperform DL on structured data.

| Method | Typical AUC | Speed | Key Advantage |
|---|---|---|---|
| XGBoost | 0.95–0.99 | Medium | Best overall tabular; built-in feature importance |
| LightGBM | 0.95–0.98 | Fastest | 3–5× faster; high-cardinality cats |
| CatBoost | 0.95–0.98 | Slow | Native categoricals; ordered boosting |
| Stacking | 0.96–0.99 | Slowest | Best combined performance |

Key findings:

- IEEE-CIS Fraud Detection: all top-10 used XGBoost/LightGBM. Winning AUC ∼0.966.
- CatBoost advantage disappears with manual target encoding (Oliveira et al., 2021).
- LightGBM within 0.5% AUC of XGBoost but 3–5× faster training—matters for production retraining.
- Lokanan et al. (2024, *J. Financial Crime*): RF (AUC 0.94) and XGBoost (0.95) match or beat LSTM (0.93) on tabular AML data.
- Stacking (RF + XGBoost + LightGBM + LR meta-learner) improves AUC by 0.5–1.5% over best single model.

### 3.2 Unsupervised Methods

| Method | AUC | Strengths | Weaknesses |
|---|---|---|---|
| Isolation Forest | 0.78–0.85 | No labels; fast; scalable | Lower precision standalone |
| Extended IF | +1–2% | Better correlated features | – |
| LOF | 0.72–0.80 | Local anomalies | Doesn't scale; k-sensitive |
| One-Class SVM | 0.70–0.80 | Only needs normal class | Kernel choice; doesn't scale |
| DBSCAN | 0.60–0.75 | Structural outliers | Low precision |
| PCA | 0.70–0.85 | Fast; interpretable | Misses nonlinear |

Comparative ranking (Hilal et al., 2022): IF > Autoencoder > LOF > OCSVM > DBSCAN.

**Key insight.** Unsupervised methods are best as *feature extractors*, not standalone detectors. Two-stage approach: IF anomaly scores as features for supervised model improves AUC by ∼3% (Jullum et al., 2020). This "anomaly score as feature" paradigm is now standard.

### 3.3 Hybrid Rule + ML

Industry standard. Pure ML replacement of rules is rare due to regulatory requirements:

- **Three-tier FRAML architecture** (2023): (1) hard rules for regulatory compliance, (2) soft rules for known typologies, (3) ML for emerging patterns.
- Rocha-Salazar et al. (2021): Rules generate alerts; RF re-scores. Alert volume −70%; SAR precision from 2% (rules) to 15% (hybrid).
- Soltani Delgosha and Hajiheydari (2021): Staged rule+ML: −40% false positives while maintaining recall.
- Regulators mandate explainable triggers ⇒ rules for audit trail, ML for ranking.

## 4 Contextual Anomaly Detection

**Core idea.** A $50K wire is normal for a corporate treasury but anomalous for a student account. Contextual anomaly detection models $P(\mathbf{y}|\mathbf{c})$—the expected *behavioral* features $\mathbf{y}$ given *context* features $\mathbf{c}$—rather than a global joint distribution $P(\mathbf{x})$. An observation is contextually anomalous if its behavior deviates from what is expected *given its context*, even if it appears normal globally.

This is a relatively thin literature compared to global anomaly detection, but it is the most operationally relevant framing for AML.

## 4.1 Foundational Methods

| Method | Year | Key Idea | Venue |
|---|---|---|---|
| CAD | 2007 | Foundational: model $P(\mathbf{y}\|\mathbf{c})$ via conditional Gaussian mixture; anomaly = low conditional likelihood | IEEE TKDE |
| ROCOD | 2016 | Local (KNN-based) + global (regression) models with adaptive weighting by context density; 40× speedup | CIKM |
| QCAD | 2023 | Quantile regression forests for conditional quantiles; explainable via quantile intervals | DAMI |
| ConQuest | 2024 | Auto-discovers multiple useful contexts via genetic algorithm; MCAF multi-context scoring | IJDSA 2025 |

**Song et al. (2007, IEEE TKDE).** The foundational "conditional anomaly detection" paper. Splits features into context $\mathbf{c}$ and behavioral $\mathbf{y}$, models $P(\mathbf{y}|\mathbf{c})$ with conditional Gaussian mixtures. Anomaly score = negative log conditional likelihood. Showed contextual detection finds anomalies invisible to global methods.

**ROCOD** (Liang and Parthasarathy, 2016, CIKM). Addresses context sparsity: when few peers exist, the local model (KNN-based expected behavior) is unreliable. ROCOD blends a local peer model with a global regression model via adaptive weighting—local dominates in dense regions, global in sparse. 40× faster than prior contextual methods. The local model estimates expected behavior from contextual neighbors; the global model learns overall context→behavior dependencies.

**QCAD** (Li and van Leeuwen, 2023, *Data Mining and Knowledge Discovery* 37(6):2517–2563). Key advancement over ROCOD: uses *quantile regression forests* to model $P(\mathbf{y}|\mathbf{c})$ instead of the conditional mean. The quantile-based approach captures the full conditional distribution—wider quantile intervals indicate lower density and higher anomaly likelihood. Anomaly scores aggregated over behavioral features. Explainable: quantile intervals directly show why a point is anomalous ("value falls outside the 1st–99th percentile given this context"). Outperforms ROCOD on both accuracy and interpretability. Open-source: `github.com/ZhongLIFR/QCAD`.

**ConQuest** (Calikus, Nowaczyk, and Dikmen, 2024, *Int. J. Data Science and Analytics* 19:99–113). Solves the *context discovery* problem: which features should be context vs. behavioral? Uses a multi-objective genetic algorithm with three objectives: (1) maximize dependency between context and behavior, (2) minimize contextual redundancy, (3) maximize anomaly discrimination. Produces multiple useful contexts, each scored via a novel Multi-Context Anomaly Factor (MCAF). Outperforms SOTA on 25 datasets. Directly interpretable: each context provides a different "reason" for anomalousness.

## 4.2 Deep Contextual Methods

### 4.2.1 Conditional Wasserstein Autoencoder (CWAE)

**King et al. (2025), "Contextual Learning for Anomaly Detection in Tabular Data"** (arXiv:2509.09030, submitted to TMLR). This is the most directly relevant recent paper for deep

contextual AD on tabular data.

**Architecture.** Splits features into context $\mathbf{c}$ and content/behavioral $\mathbf{y}$. Deterministic encoder $\mathbf{z} = E_\phi([\mathbf{y}; \mathbf{c}])$, decoder $\hat{\mathbf{y}} = D_\theta([\mathbf{z}; \mathbf{c}])$ reconstructs behavioral features only. Loss:

$$\mathcal{L} = \sum_{i=1}^{d} \mathrm{CE}(y_i, \hat{y}_i) + \lambda \cdot \mathrm{MMD}^2(P(\mathbf{z}), Q(\mathbf{z}|\mathbf{x}))$$

Uses MMD with multi-scale IMQ kernel instead of KL divergence $\Rightarrow$ deterministic encoder, no sampling noise, bounded gradients.

**Context selection.** Bilevel optimization: outer loop iterates over candidate context features; inner loop trains CWAE for 1 epoch; selects context minimizing joint validation loss $\mathcal{L}_{\mathrm{val}} = -\log P(\mathbf{y}, \mathbf{c})$. Leverages the finding that early-trained models proxy generalization quality.

**Theoretical justification.** Variance decomposition: $\mathrm{Var}(\mathbf{y}) = \mathbb{E}_{\mathbf{c}}[\mathrm{Var}(\mathbf{y}|\mathbf{c})] + \mathrm{Var}_{\mathbf{c}}(\mathbb{E}[\mathbf{y}|\mathbf{c}])$. A context-conditional detector only needs to model $\mathrm{Var}(\mathbf{y}|\mathbf{c})$, which is typically much smaller than total $\mathrm{Var}(\mathbf{y})$ $\Rightarrow$ higher SNR for anomaly detection.

**Results** on 8 benchmarks (Bank, Beth, Census, CMC, KDD, LANL, SF, Spotify):

| Method | Avg AUC | Avg Rank | Type |
|---|---|---|---|
| CWAE (proposed) | **0.797** | **2.75** | Contextual deep |
| WAE (non-contextual) | 0.760 | 4.25 | Global deep |
| CVAE | 0.745* | – | Contextual deep |
| CMDN | 0.756* | – | Contextual deep |
| CADI | 0.734* | – | Contextual tree |
| RCA | 0.769 | 3.38 | Global deep |
| DIF | 0.758 | 3.50 | Global tree |
| SLAD | 0.695 | 5.88 | Global deep |

*Estimated from ablation tables; exact ordering varies by dataset.

Key finding: with oracle context selection, CWAE achieves avg AUC 0.839 (rank 1 on 6/8 datasets). The gap between automatic and oracle context selection highlights that *context feature choice* is the critical design decision.

**CWAE vs. CVAE comparison.** CWAE outperforms CVAE (+11.69% avg improvement over non-contextual baseline vs. CVAE's +6.21%). CWAE's deterministic encoder avoids CVAE's numerical instability: CVAE requires $\log \sigma^2$ clamping to prevent divergence on skewed financial data, and Monte Carlo sampling adds inference noise.

### 4.2.2 Conditional VAE (CVAE) for Anomaly Detection

CVAE models $q_\phi(\mathbf{z}|\mathbf{x}, \mathbf{c})$ with KL regularization. Applied to contextual AD by conditioning the generative model on context features. Three instability sources on financial data: (1) $\log \sigma^2 \to -\infty$ when encoder predicts near-zero variance, (2) $\log \sigma^2 \to +\infty$ causes overflow, (3) reparameterization amplifies noise when $\sigma$ is extreme. The KL term also forces a "fuzzy" Gaussian latent space that may reconstruct anomalies too well.

**Disentangled CVAE** (IEEE 2024): Combines $\beta$-VAE, CVAE, and total correlation to disentangle context from anomaly signals. Addresses information loss while incorporating known sources of variation.

### 4.2.3 Other Deep Contextual Methods

- **CMDN** (Conditional Mixture Density Network): Neural network-parameterized Gaussian mixture for $P(\mathbf{y}|\mathbf{c})$. More flexible than single Gaussian but harder to train. +7.47% improvement over non-contextual (King et al., 2025).
- **CADI** (Contextual Anomaly Detection using Isolation Forest): Extends IF with density-aware splits conditioned on context. +4.18% improvement. Tree-based ⇒ more interpretable.

## 4.3 Peer Group and Peer-Normalized Methods

The operational form of contextual AD in AML: compare each entity to its "peers" rather than the global population.

**Bolton and Hand (2001).** Foundational peer group analysis for fraud detection. Group entities by context features, flag deviations from peer behavior. The core idea: compute per-entity statistics relative to peer group mean and standard deviation.

**Peer-normalized Z-score approach.** For each entity $i$ with context $\mathbf{c}_i$ and behavioral feature $y_i$:

$$z_i = \frac{y_i - \mu_{\text{peer}}(\mathbf{c}_i)}{\sigma_{\text{peer}}(\mathbf{c}_i)}$$

where $\mu_{\text{peer}}$ and $\sigma_{\text{peer}}$ are estimated from context-similar entities (KNN with kernel weighting, or hard peer group assignment). This is effectively the simplest form of contextual AD.

**Peer-Normalized Isolation Forest.** Apply IF to peer-normalized features rather than raw features. This operationalizes contextual AD without changing the detector—only the feature representation changes. The normalization removes context-induced location/scale shifts, exposing within-context anomalies. Computationally cheap: no model retraining needed per context.

**Uncertainty-Aware Contextual AD** (arXiv:2507.04490, 2025). Extends the peer-normalized Z-score by treating it as a *random variable* with confidence intervals. Uses two heteroscedastic Gaussian process regression models—one for the conditional mean $\mathbb{E}[y|\mathbf{c}]$, one for the conditional variance $\text{Var}(y|\mathbf{c})$. Provides calibrated uncertainty: wide confidence intervals ⇒ low reliability of anomaly assessment. Outperforms QCAD and ROCOD on benchmark datasets and a real-world cardiology application. Key advantage: enables adaptive, uncertainty-driven decision-making rather than hard thresholds.

## 4.4 When Does Context Help?

Not always. Contextual methods add value specifically when anomalies are *contextually unusual but globally normal*:

| Anomaly Type | Example | Best Detector |
|---|---|---|
| Global | $10M single transfer (unusual for anyone) | Standard IF, XGBoost |
| Contextual | $50K wire from a student account (normal for corp.) | Peer-normalized IF, CWAE, QCAD |
| Structural | Fan-out → fan-in chain (smurfing) | GNN, FlowScope |
| Temporal | Dormant account suddenly active | LSTM, TGN |

Key empirical finding: the magnitude of improvement from contextual methods depends on (1) the *type* of anomaly present and (2) the contamination rate. Under context-mismatch contamination, contextual methods can substantially outperform global methods (e.g., 39.8% vs. 20.8% precision at top-5% in one controlled experiment). Under global contamination, the advantage disappears or reverses.

**Diagnostic heuristic.** Low agreement between a global detector (IF) and a contextual detector (peer-normalized IF) at a given threshold can indicate the presence of contextual structure in the data. This can inform method selection before committing to a full pipeline.

### 4.5 KYC Integration in Practice

- **Customer 360 Profiles**: Dynamic integration of KYC data, transaction history, external datasets for trend/anomaly identification and peer group comparison.
- **HSBC Dynamic Risk Assessment**: ML models on transaction patterns + network behaviors + KYC data. Result: **60% alert reduction, 2–4× true positive increase**.
- **AI risk rating**: Incorporating behavioral and transactional factors into dynamic customer risk scores, enabling peer group comparisons.
- Customer segmentation (k-means, DBSCAN) enables contextual thresholding: entities far from cluster centroid get heightened scrutiny.

### 4.6 Entity Resolution

- **Knowledge graphs**: Connect people, companies, accounts, transactions. Trace ownership chains across borders/shells. Link duplicate records.
- Data-driven name matching: 90% more accurate than legacy fuzzy matching (handles transliterations, scripts, name variants).
- Graph-based entity resolution (Neo4j) for shared addresses, phones, IPs.
- Composite AI: entity resolution + graph analytics for proactive discovery.

### 4.7 Landscape Assessment

**The honest summary.** Contextual anomaly detection is a relatively thin literature:

- The core lineage is Song (2007) → ROCOD (2016) → QCAD (2023) → ConQuest (2024) → CWAE (2025).
- Most AML papers do global anomaly detection. Very few model $P(\mathbf{y}|\mathbf{c})$ explicitly.
- The deep contextual methods (CWAE, CVAE, CMDN) are recent (2024–2025) and largely untested on AML-specific data. King et al. (2025) benchmark on general tabular datasets (Bank, Census, KDD), not AML transaction data.
- Peer group analysis is widely used in industry AML (Bolton and Hand, 2001) but rarely published as formal ML research—it sits in the practitioner/vendor space.
- The context feature selection problem (which features are context vs. behavioral?) is largely unsolved. ConQuest's genetic algorithm and King et al.'s bilevel optimization are the only principled approaches.
- **Gap**: No published work combines CWAE/WAE-MMD with AML-specific evaluation on transaction monitoring data with realistic laundering typologies. Similarly, peer-normalized IF with systematic evaluation across injection strategies is understudied.

# 5 Large-Scale Entity Resolution

Entity resolution (ER)—also called record linkage, deduplication, or identity resolution—is the task of determining which records across (or within) datasets refer to the same real-world entity. At AML scale (100M+ clients across multiple systems), naive pairwise comparison is $O(n^2)$ and infeasible. This section covers blocking strategies, matching methods, and scalable architectures.

## 5.1 The Blocking Problem

**Blocking is 10× more important than matching.** Reducing candidates from $n^2$ to $0.01 \times n^2$ makes everything tractable. Modern blocking approaches:

| Approach | Mechanism | Scale |
|---|---|---|
| Standard blocking | Hash on key attributes (name initial + DOB) | Simple; misses typos/variants |
| Sorted neighborhood | Sort by blocking key; slide window | Better recall; still key-dependent |
| LSH (Locality Sensitive Hashing) | Hash similar items to same bucket with high probability | Scalable; tunable recall/precision |
| ANN (Approximate Nearest Neighbor) | Embedding → FAISS/HNSW index → nearest neighbors | State-of-the-art; semantic similarity |
| Neural LSH | Learned hash functions via deep networks | Up to 6× faster index, 2× faster query (DET-LSH) |

**Neural LSH for Entity Blocking** (Amazon, 2024, arXiv:2401.18064). Proposes training deep neural networks as hashing functions for complex metrics. Significantly outperforms traditional LSH on entity resolution benchmarks.

**BlockingPy** (2025, arXiv:2504.04266). Open-source package implementing ANN blocking via FAISS (exact KNN, LSH, HNSW). Designed for entity resolution at scale with imperfect data.

**Hashed Dynamic Blocking.** Applied to a 530M-row industrial dataset, detecting 68 billion candidate pairs in under three hours.

## 5.2 Matching Methods

### 5.2.1 Probabilistic: Fellegi-Sunter Model

The classical approach. For each record pair, compute match weights from field-level agreement/disagreement probabilities:

$$w_j = \log_2 \frac{P(\text{agree}_j|\text{match})}{P(\text{agree}_j|\text{non-match})}$$

Aggregate weights across fields; threshold for match/non-match/clerical review.

**Splink** (UK Ministry of Justice, open-source). The leading implementation of Fellegi-Sunter at scale. Key capabilities:

- Links 100M+ records using Spark or AWS Athena backends; 1M records on laptop in ∼1 minute.
- Unsupervised (no labeled training data): learns match/non-match weights via EM algorithm.

- Won Civil Service Awards 2025 (Innovation) and OpenUK Awards 2025.
- Used by Harvard Medical School on 8.1M records (American Journal of Epidemiology, 2025).
- Supports fuzzy string comparisons (Jaro-Winkler, Levenshtein), date comparisons, geographic distance.

#### 5.2.2 Deep Learning: Transformer-Based Matching

The paradigm shift since 2020: use pre-trained language models (BERT, SentenceBERT, DistilBERT, RoBERTa) to encode record pairs into embeddings and classify match/non-match.

**Ditto** (Li et al., 2020). Landmark paper: BERT-based entity matching achieving SOTA, beating previous benchmarks by up to 29% F1. Key idea: serialize record pairs as text, fine-tune BERT as binary classifier. Introduced data augmentation for ER (attribute-level).

**SentenceBERT for ER.** SentenceBERT models rank first on most ER benchmarks because they natively produce embedding vectors suitable for similarity search. Used both for blocking (embed $\rightarrow$ ANN search) and matching.

**EMBA** (2024, EDBT). Multi-task learning of BERT for entity matching, improving over single-task fine-tuning.

**Ensemble transformers** (2024). Combining multiple transformer models for better entity matching through ensembling.

**CSGAT** (2025, *Scientific Reports*). Contextual Semantics Graph Attention Network extracting contextual information at token and attribute levels to generate semantically fused embeddings for entity matching.

#### 5.2.3 LLM-Based Entity Resolution

The newest frontier (2024–2025):

- **Peeters and Bizer (2024)**: GPT-4 outperforms fine-tuned pre-trained language models by **40–68%** on unseen entity types in zero-shot settings. No training data required.
- **GPT-4o**: More cost-effective alternative; slightly lower performance than GPT-4.
- **In-context clustering ER** (2025, arXiv:2506.02509): Packs multiple records into sets for direct in-context clustering via LLM. Generalizes SOTA approaches with improved cost-effectiveness.
- **Balancing efficiency and quality** (Springer 2025): Explores optimal prompting strategies for LLM-based ER on structured data. "Selecting" strategy identified as most cost-effective.
- **Limitation**: Cost at 100M+ scale—each LLM call is expensive. LLMs best used for high-uncertainty pairs after initial blocking+probabilistic matching.

### 5.3 Scalable Architectures and Tools

| Tool | Scale | Backend | Key Feature |
|---|---|---|---|
| Splink | 100M+ | Spark, Athena, DuckDB | Probabilistic (Fellegi-Sunter); unsupervised |
| Zingg | 100M+ | Spark, Snowflake, Databricks | ML-based blocking + matching; active learning |
| Senzing | Billions | In-memory graph | Real-time ER; entity-centric; used by banks |
| AWS ER | 100M+ | AWS Glue ML | Managed service; integrates with Zingg |
| Neo4j | 100M+ | Graph database | Graph-based ER; ownership chains |

**Zingg** (open-source, AGPL v3.0). ML-based entity resolution for Spark, Snowflake, Databricks. Learns blocking model that indexes near-similar records—typical comparisons are 0.05–1% of the problem space. Supports both probabilistic (fuzzy) and deterministic matching. Active learning: human labels a small sample, model generalizes.

**Senzing** (commercial). In-memory, entity-centric ER engine. Processes billions of records in real-time. Used in financial crime, AML, and government applications. Key differentiator: maintains a live entity graph that updates incrementally as new records arrive—no batch reprocessing.

### 5.4 ER for AML: Specific Challenges

- **Name matching across scripts/cultures**: Transliterations (Arabic ↔ Latin), name ordering (family name first vs. last), aliases, nicknames. Data-driven name matching achieves 90% more accuracy than legacy fuzzy matching.
- **Shell companies and beneficial ownership**: Resolving entities behind layers of shell companies across jurisdictions. Knowledge graphs trace ownership chains; graph-based ER links shared addresses, phones, directors.
- **Real-time vs. batch**: AML requires both—batch for initial deduplication/linking across 100M+ records; real-time for new customer onboarding and ongoing monitoring.
- **Privacy constraints**: Cross-institution ER without sharing PII. Privacy-preserving record linkage (PPRL) uses encrypted/hashed tokens for comparison.
- **Scale progression**: <1M records: DuckDB/Python. 1–100M: Splink/Zingg on Spark. 100M+: Senzing, custom distributed architectures, or graph databases (Neo4j).

### 5.5 ER Performance Benchmarks

**Leipzig benchmark datasets** (Database Group Leipzig): Standard ER benchmarks with ground truth, including product, bibliographic, and geographic datasets at various scales.

**Key results on benchmarks:**

- BERT/Ditto (2020): +29% F1 over previous SOTA on structured ER benchmarks.
- SentenceBERT: Ranks 1st on most datasets; never below 4th.
- GPT-4 (zero-shot): +40–68% over fine-tuned PLMs on unseen entity types.
- Splink: Near-SOTA accuracy at 100M+ scale with zero training data.

**The emerging architecture for 100M+ AML ER:**

1. **Embedding**: SentenceBERT or domain-specific encoder → vector per record
2. **Blocking**: ANN index (FAISS/HNSW) → candidate pairs (0.01–1% of $n^2$)
3. **Matching**: Probabilistic (Splink) or fine-tuned transformer
4. **Uncertain pairs**: LLM (GPT-4) for high-ambiguity cases
5. **Graph consolidation**: Neo4j/knowledge graph for entity-level view

# 6 Feature Engineering

**Feature engineering matters more than model choice.** The gap between well-engineered XGBoost and poorly-engineered DL is larger than between model architectures.

| Category | Examples | Impact |
|---|---|---|
| Velocity | Txn count/amount in 1h, 6h, 24h, 7d, 30d windows; unique counterparties; frequency acceleration; time since last txn | AUC +5–8% (Bahnsen et al., 2016) |
| Behavioral | Amount ratio to hist. mean; Z-score; new merchant/country; dormancy; day/time deviation | Most important category |
| Network/Graph | Degree; PageRank; clustering coeff.; fan-in/fan-out; cycle detection; community membership | F1 +7% even without GNNs (Alarab et al., 2020) |
| Aggregation | Per-customer stats (mean, std, skew); per-merchant fraud rate; per-country-pair amounts; per-channel stats | Foundation for behavioral deviation |
| Temporal | Day of week; hour; weekend/holiday; days since account open; seasonal patterns | Laundering timing patterns |
| Risk Indicators | FATF country risk; customer risk rating; PEP; industry risk; historical SAR count | Contextual conditioning |

**Two-stage feature architecture.** Gaining popularity:

1. Unsupervised scoring (IF anomaly scores, PCA reconstruction errors, autoencoder loss, cluster distances)
2. Supervised classification on original features + Stage 1 scores

Addresses label scarcity. Jullum et al. (2020): ∼3% AUC improvement.

## 6.1 Handling Multicollinearity in Feature-Rich AML Pipelines

Feature engineering for AML produces dozens to hundreds of correlated features (e.g., txn count 24h ↔ txn count 7d; amount mean ↔ amount median). This multicollinearity does not degrade *prediction* in tree-based models but causes three practical problems: (1) unstable/misleading feature importance, (2) inflated model complexity, and (3) attribution dilution across correlated features in post-hoc explanation methods—undermining explainability.

**Detection.**

- **Pearson / Spearman correlation matrix**: Drop one of any pair with $|r| > 0.90$–$0.95$. Simple, interpretable, but only captures pairwise linear relationships.
- **Variance Inflation Factor (VIF)**: Regresses each feature against all others. VIF $> 5$–$10$ indicates problematic collinearity. More comprehensive than pairwise correlation but expensive at high dimensionality ($O(p^2 n)$).
- **Eigenvalue analysis**: Small eigenvalues of the feature covariance matrix indicate near-linear dependencies. Condition number $> 30$ signals severe multicollinearity.

**Mitigation strategies.**

| Method | How It Works | When to Use |
|---|---|---|
| Correlation filter | Drop one of each highly correlated pair ($\|r\| > 0.9$) | Quick first pass; interpretable |
| VIF-based removal | Iteratively remove highest-VIF feature until all VIF $< 5$ | Linear models, logistic regression |
| Variable clustering (VARCLUS) | Hierarchical clustering of features; keep one representative per cluster | Large feature sets; preserves coverage of feature space |
| PCA / kernel PCA | Project onto orthogonal components | When interpretability of individual features is not required |
| Mutual information (mRMR) | Select features maximizing relevance to target while minimizing redundancy among selected features | Supervised setting with labels; captures nonlinear dependencies |
| LASSO / Elastic Net | $L_1$ penalty drives correlated feature coefficients to zero | Embedded selection; regularization path shows stability |
| Recursive Feature Elimination (RFE) | Iteratively remove least-important features using model importance | Works with any model; computationally expensive |
| Attribution-based selection | Cluster features by explanation-value correlation; keep representative of each cluster | Post-hoc; directly addresses attribution dilution |

**Practical notes for AML feature engineering:**

- **Tree models (XGBoost/LightGBM) are robust to multicollinearity for prediction**—model accuracy is minimally affected. However, feature importance and post-hoc explanations become unreliable when features are correlated (importance "splits" across redundant features). Since explainability is required for FINTRAC, this must be addressed.
- **Recommended pipeline**: (1) Correlation filter ($|r| > 0.9$) as first pass, (2) group remaining features by semantic category (velocity, behavioral, network, risk), (3) within each group, apply VIF or mRMR to select non-redundant representatives, (4) validate with attribution clustering post-training.
- **Domain-informed grouping**: Rather than purely statistical removal, group by business meaning. E.g., keep `txn_count_24h` and `txn_count_30d` (different time horizons) but drop `txn_count_23h` (redundant with 24h). Domain knowledge prevents removing features that are statistically correlated but semantically distinct.
- **Interaction analysis**: For pairs that survive filtering, interaction values (e.g., SHAP $\phi_{ij}$ or permutation-based methods) can reveal whether both features contribute *independently* to predictions. If interaction is near-zero and main effects are similar, one can be dropped.

- **Stability selection** (Meinshausen & Bühlmann, 2010): Run feature selection (LASSO, RFE) on random subsamples; keep features selected in $> 60\%$ of runs. Robust to collinearity-induced instability.

# 7 Datasets and Evaluation

## 7.1 Public Datasets

| Dataset | Type | Size | Imbal. | Notes |
|---|---|---|---|---|
| Elliptic | Real BTC | 203K txns | 2% | Most-used; 49 timesteps; 166 features |
| Elliptic++ | Real BTC | 204K+ | 2% | +address-level graph; subgraph labels |
| AMLSim (IBM) | Synthetic | Config. | Config. | Open-source; controllable patterns |
| PaySim | Synthetic | 6.3M txns | 0.13% | Mobile money; network structure |
| IBM AML Synth. | Synthetic | 180M+ | Config. | Large-scale multi-bank |
| IEEE-CIS | Real CC | 590K | 3.5% | Kaggle competition; rich features |
| European CC | Real CC | 284K | 0.17% | PCA-anonymized |

**Synthetic data.** Altman et al. (2023, *Nature Scientific Data* / NeurIPS 2023): large-scale synthetic AML benchmark. Market projected \$310–576M (2024) → \$1.8–16.7B by 2030. By 2025, 75% of large banks use synthetic data for AI. Tools: CTGAN, Copula GAN, TVAE—96–99% utility equivalence.

## 7.2 Class Imbalance

AML: 0.1–2% positive rate. Approaches:

- **SMOTE**: Most widely used. RF+SMOTE: F1 from 0.72 → 0.87 (Al-Hashedi and Magalingam, 2021).
- **SMOTE-GAN hybrids**: SMOTified-GAN, etc.
- **Cost-sensitive learning**: Class weights, focal loss.
- **GraphSMOTE**: Graph-aware oversampling.
- **AMLNet** (2025): SMOTE + RandomUnderSampler + multi-agent framework.

## 7.3 Metrics

AUC alone is insufficient for AML:

- **F1 (illicit)**: More discriminating on imbalanced data. Ranges 0.47 (RF) to 0.88 (PC-GNN).
- **Precision@top-$k$**: How many top-$k$ alerts are true? Maps to analyst workload.
- **Alert-level precision**: % of alerts → SARs. Rules: ∼2%; hybrid ML: ∼15%.
- **FP reduction**: Primary industry KPI. ML consistently 40–60% over rules.
- **PSI**: Monitors concept drift; triggers retraining.

### 7.4 Concept Drift

Launderers adapt ⇒ model degradation:

- Deprez et al. (2025): Continual graph learning review—replay, regularization, architecture strategies.
- Cross-regional continual learning with knowledge transfer (TKDE 2024).
- PSI-based drift detection standard in production.
- Traditional models suffer catastrophic forgetting on fine-tuning.

## 8 Emerging Paradigms

### 8.1 Federated Learning

Banks can't share data but laundering spans institutions:

- **FAML**: Multi-bank collaborative training without data sharing. Up to **30% higher accuracy** vs. isolated.
- **FedGraph-VASP** (2026): Federated graph learning + post-quantum crypto for cross-institutional AML.
- One implementation on PaySim: 0.999 accuracy, 0.1% false alarm rate.
- **FHE + GNNs**: Fully homomorphic encryption for privacy-preserving graph ML.
- **IBM Research**: Privacy-preserving federated learning for financial crime.
- **Federated Graph Anomaly Detection via Disentangled Representations** (TheWebConf 2025).

### 8.2 Explainability

Regulatory non-negotiable (FinCEN, FCA, EBA, FATF):

- **SHAP/TreeSHAP**: Widely used. Jullum et al. (2020) reported analysts 30% faster with SHAP explanations. However, SHAP values can be unstable with correlated features (common in AML), and computational cost grows with feature count. Should be used alongside—not as a replacement for—domain expertise.
- **LIME**: An alternative to SHAP for local explanations, though less stable on correlated tabular features. Useful for quick per-alert explanations but sensitive to perturbation strategy.
- **SEFraud** (KDD 2024): Self-explainable fraud via interpretative mask learning.
- Counterfactual explanations ("this alert would not have fired if...") demanded by regulators.
- Global surrogates (decision trees/GAMs approximating black-box) for regulatory documentation.
- EU AML Directive (2020), FinCEN guidance (2021): mandate "understandable" automated detection.

### 8.3 Agentic AI

Most recent frontier (2025–2026):

- **Nasdaq Verafin**: Agentic AI Workforce automating low-value, high-volume compliance. Up to 50% AML workload automation.
- **Co-Investigator AI** (2025): Agentic SAR generation—faster and more accurate.
- McKinsey (2025): Agentic AI automates EDD reviews (adverse media, historic activity, escalation).

- Agent-based KYC: onboarding from weeks → hours.
- Human-in-the-loop: AI handles lower-risk; humans retain high-level decisions.

### 8.4 Adversarial Robustness

- CARE-GNN: Anti-camouflage (fraudsters connecting to benign nodes).
- RoBCtrl (2025): RL-based attacks on GNN bot detectors—highlights deployed system vulnerability.
- Adversarially robust temporal graph contrastive learning (2025).
- Graph adversarial training (2024) for GNN robustness.

## 9 Decision Framework

| Scenario | Data | Labels | Recommended Approach |
|---|---|---|---|
| Tabular, labeled | Tabular | Yes | XGBoost/LightGBM + post-hoc explanations |
| Tabular, no labels | Tabular | No | IF + two-stage |
| Graph, labeled | Network | Yes | GraphSAGE/GAT + temporal |
| Graph, no labels | Network | No | GAE or self-supervised GNN |
| Multi-institution | Any | Mixed | Federated learning (FAML) |
| Real-time | Stream | Any | TGN / streaming GNN |
| Regulatory focus | Any | Any | Hybrid rule+ML + explainability layer |
| Few labels | Any | Few | Self-supervised + fine-tuning |

## 10 Industry Adoption

| Organization | Method | Scale | Year |
|---|---|---|---|
| HSBC | Dynamic Risk Assessment (ML+KYC) | Banking | 2024 |
| Ant Group | Heterogeneous GNN | Billions of nodes | 2020+ |
| eBay | xFraud (Het-GNN) | Billions of txns | 2022 |
| PayPal | GNN + streaming | Real-time | 2021+ |
| Feedzai | Graph-enhanced ML | Banking AML | 2022+ |
| Elliptic | AI on Bitcoin | 200M txns | 2024 |
| Nasdaq Verafin | Agentic AI | Banking AML | 2025 |

**SAS/KPMG/ACAMS 2025 survey** (850 ACAMS members): 18% have AI/ML in production; 18% piloting; 25% plan in 12–18 months; 40% no plans. Top priority: FP reduction (38%). ML top tech choice (58%). Top barrier: lack of regulatory imperative (37%, overtook budget). Regulator enthusiasm declining (51% say regulators encourage, down 15pts from 2021).

## 11 Supervised Learning with STR-Labeled Data

When dispositioned STR (Suspicious Transaction Report) data is available, supervised learning becomes viable and is generally the highest-performing approach.

### 11.1 Training Data Construction

- **Label source**: Filed STRs = positive class; reviewed-and-cleared alerts = negative class. Raw non-alerted transactions are *not* true negatives (unknown positives problem).
- **Recommended window**: 6–24 months of dispositioned cases. Too short = insufficient positive examples; too long = concept drift contaminates the training set.
- **Label quality**: STR filing decisions vary by analyst. Consensus labels (multiple reviewers) or adjudicated labels (senior review) improve training signal. Label noise is the single largest risk to supervised AML models.
- **Class imbalance**: STR filing rates are typically 2–5% of alerts, which are themselves <1% of transactions. Effective mitigation: SMOTE, cost-sensitive learning ($w_{\text{pos}} = N_{\text{neg}}/N_{\text{pos}}$), or focal loss. GraphSMOTE extends oversampling to graph-structured data.

### 11.2 Supervised Model Performance

Gradient-boosted trees dominate supervised AML:

- XGBoost on tabular AML features: **AUROC 0.961** reported on production-scale labeled data (Jullum et al., 2020; DNB Bank study).
- LightGBM: comparable AUROC with 2–3× faster training; preferred at scale.
- Random Forest: strong baseline (AUROC ∼0.94) but less competitive than boosting.
- Deep learning (DNN, LSTM): marginal gains over XGBoost on tabular data; justified only when sequential or graph structure is exploited.

**False positive reduction** is the primary value proposition of supervised models over rule-based systems. Literature consistently reports **60–80% FP reduction** while maintaining or improving recall:

- Deloitte/SAS (2024): 60% FP reduction with supervised ML overlay on existing rules.
- Guidehouse (2023): ML models reduce alert volumes by 70% with no loss in STR filing rate.
- FinCEN (June 2024 proposed rule): explicitly endorses ML for "greater precision in assessing customer risk" and "reducing false positives."

### 11.3 Semi-Supervised Extensions

When labeled data is limited (common in early deployment):

- **Two-stage pipeline**: (1) unsupervised anomaly scores (IF, autoencoder, LOF) as features, (2) supervised classifier on the enriched feature set. This is the most common production architecture.
- **Self-training / pseudo-labeling**: Train initial model on labeled subset, predict on unlabeled, add high-confidence predictions to training set. Iterate.
- **Positive-unlabeled (PU) learning**: Treats non-STR transactions as unlabeled (not negative). Appropriate given the unknown-positives problem. Methods: Elkan & Noto (2008), unbiased PU risk estimator (Kiryo et al., 2017).
- **Contrastive pre-training**: Self-supervised on full transaction corpus, then fine-tune on STR labels. FLAG (KDD 2025), TREASURE (2025) demonstrate this for financial data.

# 12 Complex Behavior Detection

FINTRAC and international regulators increasingly focus on *efficacy*—the ability to detect complex, multi-step laundering typologies, not just individual suspicious transactions.

## 12.1 Laundering Typologies

- **Structuring (smurfing)**: Breaking large amounts into sub-threshold deposits across accounts/branches. Rule-based detection (aggregation thresholds) is well-established but easily evaded by varying amounts.
- **Layering**: Moving funds through multiple intermediary accounts/entities to obscure the trail. Requires multi-hop graph analysis.
- **Mule networks**: Recruiting third-party accounts to receive and forward funds. Detected via fan-in/fan-out patterns and behavioral anomalies.
- **Trade-based laundering (TBML)**: Over/under-invoicing in international trade. Requires trade data integration and price anomaly detection.
- **Nested/shell structures**: Using corporate hierarchies to obscure beneficial ownership. Entity resolution is critical here.

## 12.2 Detection Approaches by Typology

| Typology | Best Approach | Key Reference |
|---|---|---|
| Structuring | Temporal aggregation + supervised | XGBoost with rolling-window features |
| Smurfing | Subgraph mining | GARG-AML (2025): second-order neighborhood scoring; SMoTeF: temporal-constrained flow |
| Layering | Multi-hop GNN / path analysis | FlowScope (AAAI 2020): dense subgraph on flow networks |
| Mule networks | Community detection + fan patterns | Heterogeneous GNN with role-aware embeddings |
| TBML | Price anomaly + cross-dataset | Trade price deviation models + customs data |

## 12.3 Multi-Typology Supervised Models

With STR labels, a single model can learn multiple typologies simultaneously:

- **Multi-label classification**: Each STR is tagged with typology code(s). Train XGBoost/LightGBM with multi-label targets. Enables typology-specific explanations.
- **Hybrid LSTM-GraphSAGE**: Captures both sequential transaction patterns (LSTM) and network structure (GraphSAGE). Reported **95.4% accuracy** on simulated multi-typology data.
- **Hierarchical detection**: Level 1 = anomaly score (unsupervised), Level 2 = typology classifier (supervised), Level 3 = case-level aggregation. Each level adds specificity.

# 13 FINTRAC Regulatory Alignment

FINTRAC's evolving stance creates both opportunity and obligation for ML-based AML systems:

- **Efficacy mandate**: New violation for failure to ensure compliance programs are "reasonably designed, risk-based and effective" (2025). Shifts focus from checkbox compliance to demonstrable detection capability.
- **Enforcement escalation**: 20 AMPs in 2025 alone, exceeding $200M total—the largest in FINTRAC history. Inadequate transaction monitoring is a primary target.
- **FinCEN endorsement of ML**: June 2024 proposed rule explicitly endorses AI/ML for "greater precision in assessing customer risk" and "reducing false positives." Canadian regulators are expected to follow.
- **OSFI E-23 Model Risk Management** (effective May 2027): Will require model validation, ongoing monitoring, and documentation for all models used in risk management—including AML. Architecture must be designed with validation in mind from day one.
- **Explainability requirement**: Both FINTRAC and OSFI require that automated decisions be "understandable." Post-hoc explanation methods (e.g., SHAP, LIME, surrogate models) can help satisfy this, though no single method is universally sufficient. Black-box models (deep GNNs, LLMs) require additional justification.

*Note: Some citation details (exact page/volume numbers) should be verified against original sources before formal publication. Performance metrics reflect literature-reported values under specific experimental configurations.*